

Trajectory Preprocessing

Each sample contains 30 observed vessel positions and 30 future positions,

$$H = (x_1, \dots, x_{30}), Y = (y_1, \dots, y_{30})$$

where every x_t and y_t is a two-dimensional position.

The transformed representation describes the trajectory relative to the vessel's **last observed position** and **latest observed direction of motion**. Both references are obtained from the observed history H and are then used unchanged for the future Y .

The last observed position is x_{30} . Every position p is expressed relative to it as

$$p - x_{30}$$

The observation therefore ends at

$$x_{30} - x_{30} = (0, 0)$$

and future positions describe where the vessel moves relative to the point where observation ended.

Compass orientation is removed in the same way. The vessel's latest observed direction is obtained from consecutive history positions,

$$\text{delta } x_t = x_t - x_{(t-1)}, t = 2, \dots, 30$$

Starting from the end of the history and moving backward, we take the first displacement satisfying

$$||\text{delta } x_t||_2 > 10^{-8}$$

Its heading is

$$\theta = \text{atan2}(\text{delta } x_{(t,y)}, \text{delta } x_{(t,x)})$$

If no such displacement exists, $\theta = 0$.

The trajectory is rotated so this direction points along the positive x-axis:

$$p_{\text{tilde}} = R(-\theta)(p - x_{30})$$

where

$$R(\phi) = \begin{bmatrix} \cos(\phi) & -\sin(\phi) \\ \sin(\phi) & \cos(\phi) \end{bmatrix}$$

After this transformation, every trajectory is viewed from the same position and direction: the observation ends at $(0, 0)$, with the vessel facing along the positive x-axis.

The aligned positions can be used directly. Alternatively, they can be represented by how the vessel moves from one aligned point to the next. For an aligned trajectory

$$(q_1, \dots, q_{30})$$

we store

$$d_1 = q_1, d_t = q_t - q_{(t-1)}, t = 2, \dots, 30$$

The first point is kept so no positional information is lost. For the future trajectory, d_1 is the first future position relative to x_{30} ; every following value is the movement between two consecutive future positions.

This gives the three representations used by the experiment system: **raw coordinates**, **heading-aligned positions**, and **heading-aligned displacements**.

The transformation is reversible. Displacements recover aligned positions through

$$q_1 = d_1, q_t = q_{(t-1)} + d_t$$

and aligned positions return to the original coordinates through

$$p = R(\theta)q + x_{30}$$

Normalization is applied after this representation is formed and is handled separately, since ordinary trajectory models and chart-based flow models use different normalization layouts.

Specs

- Observed trajectory: **30 positions**
- Future trajectory: **30 positions**
- Position dimension: **2**